FINANCIAL CRIME COPILOT · AML ALERT TRIAGE

Cut AML alert-triage cost by an estimated $2.1M a year — without changing the rules your regulator already trusts

For the Chief Compliance Officer and the Head of the Financial Intelligence Unit of a mid-size commercial bank

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All figures in this deck come from a working prototype on synthetic data, or from the stated assumptions

THE PROBLEM TODAY

Analysts spend almost all of their time clearing alerts that turn out to be nothing

95%

of rule-based alerts are false positives (95.1% in the prototype's alert population of 2,417)

30 min

of manual triage per alert, in arrival order, no matter how likely it is to matter (value-case assumption)

2 hrs

to write each Suspicious Activity Report (SAR), with quality that varies by analyst (value-case assumption)

For the CCO: rising labor cost, a growing backlog and regulatory SLA risk. Adding analysts scales cost linearly.

For the Head of FIU: genuinely suspicious cases wait in the same queue as noise; experienced investigators spend hours on paperwork.

WHAT CHANGES

Keep every rule and every alert — change which alerts analysts work first, and how fast a SAR gets written

Prioritized queue with reasons

A model scores each alert by how likely it is to be genuinely suspicious. Analysts work top-down; each case shows the three factors that ranked it high.

SAR draft in seconds, filed by a human

For every high-priority case a regulator-style SAR draft is ready. The analyst reviews, edits and files it — nothing is filed automatically.

One cockpit for the analyst

Queue, case detail, SAR editor and disposition buttons in a single app; every decision is recorded with who made it and when.

Answers in plain English for managers

"How many high-priority alerts are still open?" — answered from the same governed data the analysts work on, with the query shown.

THE VALUE

≈ $2.1M per year in analyst capacity, from two levers

Lever	Assumption	Hours / yr	Value / yr
Alert triage	100k alerts × 30 min = 50k hrs; **40%** of that effort removed by working the prioritized queue	20,000	$1.5M
SAR drafting	5k SARs; writing ~2 hrs → reviewing a draft ~20 min	8,333	$0.6M
Total	**Loaded analyst cost $75 / hour**	**28,333**	**≈ $2.1M**

Why 40% is conservative: in the prototype, reviewing only the top 20% of the ranked queue already caught 96.7% of the suspicious cases (next slide) — so 80% of alerts could wait. We count only a 40% cut in triage effort.

Not counted: lower regulatory-fine exposure, faster alert-to-SAR cycle time, consistent SAR structure. Your volumes and costs replace these assumptions in the business case.

PROOF FROM THE WORKING PROTOTYPE

On held-out alerts the model put almost every suspicious case at the top of the queue

96.7%

of suspicious cases caught by reviewing only the top 20% of the queue (29 of 30, held-out test set of 605 alerts)

0.985

ROC-AUC on the held-out set (PR-AUC 0.894) — against a 4.9% base rate of suspicious alerts

121 / 121

high-priority cases with a SAR draft and a one-line rationale ready for the analyst

Review the top …	10%	20%	30%	50%
… and you catch this share of suspicious cases	93.3%	96.7%	96.7%	100%

Synthetic data with known laundering typologies (structuring, rapid movement, high-risk-country wires, PEP activity). A pilot on your alert history confirms the numbers before any rollout decision.

HOW THE ANALYST WORKS

From alert to filed SAR in one place

1 · Open the queue
Alerts ranked by priority; the highest-risk case is first, whenever it arrived.

→

2 · See why
"High-risk-country amount, number of counterparty countries, amount volatility."

→

3 · Review the draft
Edit the AI-drafted SAR narrative instead of starting from a blank page.

→

4 · Decide
File SAR, escalate, or close as false positive — recorded instantly.

→

5 · Managers ask
Open cases, SARs filed by rule, false-positive rates — in plain English.

KPIs this moves: cost per alert · alerts handled per analyst per day · alert-to-SAR cycle time · SAR quality consistency · regulatory SLA adherence

WHY IT IS SAFE TO ADOPT

Designed for the regulator's questions, not around them

Question you will be asked	How the design answers it
Could the model hide a true positive?	No alert is removed or auto-closed. The rules still raise every alert; the model only sets the order.
Who is accountable for a SAR?	The analyst. AI drafts; a person reviews, edits and files.
Can you explain a ranking to an examiner?	Every score ships with its top contributing factors, shown on the case screen.
Who can see customer PII?	National IDs and e-mails are masked centrally for anyone outside the authorized analyst group.
Is there an audit trail?	Every disposition is stored with analyst and timestamp, and lands in the governed analytics layer.

FOR THE ARCHITECT

One governed data flow on Databricks, not six tools

Lakeflow	Declarative pipeline ingests raw files: bronze → silver (data-quality expectations) → gold features
Unity Catalog	Column masks, PII tags, governed views, grants and lineage across every table below
ML	Alert-priority model tracked in MLflow, registered in Unity Catalog, SHAP explanations per alert
Gen AI	SAR drafts and rationale generated in SQL with a Foundation Model API (ai_query)
Lakebase	Postgres serving: synced case queue for the app; analyst dispositions written transactionally, then landed back in Unity Catalog
Genie Room	Natural-language questions over the queue, dispositions and KPIs — also available inside the app
Databricks App	Analyst case cockpit on the platform: reads and writes through governed resources, and records the signed-in analyst on every disposition

Code, configuration and execution logs for every layer are in the repository.

PROPOSED PATH TO VALUE

Prove it on your own alert history before anything changes for analysts

Step 1

Back-test

Score 12 months of your closed alerts. Measure how many confirmed SARs fall in the top of the queue.

Step 2

Shadow mode

Run alongside today's process; analysts keep working as they do, we compare rankings and draft quality.

Step 3

One team live

One analyst team works the prioritized queue; track cost per alert and cycle time against the baseline.

Decision we are asking for: approve the back-test on historical alerts, with your model-risk and compliance teams defining the success threshold up front.